

CLASS-8

LESSON-2 LINEAR EQUATIONS IN ONE VARIABLE

(This PDF Based on NCERT Books)

Linear equation in one variable (एक चर वाला रैखिक समीकरण), A linear equation in one variable is just a simple **balancing puzzle** with a single missing number.

The Simple Idea

- **Equation** means it has an **equals sign (=)**, like a balanced scale. The two sides must weigh the same.
- **One Variable** means there is **only one type of mystery number**, usually called **x**.
- **Linear** means the mystery number **x is simple**; it's not squared X^2 and cubed X^3 .

The Goal

Your job is to **find the value of x** that makes the scale balance.

SOLVING EQUATIONS HAVING ONE VARIABLE IN BOTH SIDES(दोनों पक्षों में एक चर वाले समीकरणों को हल करना)-

The Steps:

1. Get Variables Together

- Choose the side you want the variable to be on (usually the side where the variable is **bigger** to avoid negative numbers, but either side works).
- Use **opposite operations** (add or subtract) to move the variable term from one side to the other.
- **Crucial Rule:** Whatever you add or subtract from one side, you **must** do to the other side.

2. Get Numbers Together

- Now you have a simple equation (like $ax + b = c$).
- Use **opposite operations** (add or subtract) to move all the plain numbers away from the variable term.

3. Solve for the Variable

- Use **division** (the opposite of multiplication) to get the variable completely alone.

Example in Action

Let's solve the equation: $5x + 2 = 3x + 10$

Step	Action	Equation	Notes
1. Get Variables	Subtract $3x$ from both sides to	$5x - 3x + 2 =$	$5x - 3x = 2x$. The $3x$'s

Step	Action	Equation	Notes
Together	move it from the right.	$3x - 3x + 10$	cancel on the right.
	Simplify	$2x + 2 = 10$	basic two-step equation.
2. Get Numbers Together	Subtract 2 from both sides to move it away from the $2x$.	$2x + 2 - 2 = 10 - 2$	The 2's cancel on the left.
	Simplify	$2x = 8$	
3. Solve for x	Divide by 2 on both sides to get x alone.	$\frac{2x}{2} = \frac{8}{2}$	
	Final Answer	$x = 4$	

If you plug $x=4$ back into the original equation, both sides will equal 22 ($5(4) + 2 = 22$ and $3(4) + 10 = 22$). It's balanced!

REDUCING EQUATIONS IN SIMPLER FORM(सरल रूप में समीकरणों को काम करना)-

1. Get Rid of Parentheses (Use the Distributive Property)

If you see something like $4(x + 2)$, it means you need to multiply the number outside the parentheses (the 4) by **EVERYTHING** inside.

- **Original:** $4(x + 2) = 15$
- **Reduced:** $4x + 8 = 15$

This makes the equation much simpler to work with.

2. Combine Matching Terms (Tidying Up)

You can only add or subtract things that are the **same type**.

- **Combine Variables:** Put all the 'x' terms together.
- **Combine Constants:** Put all the plain numbers together.

Example: $7x + 5 - 3x + 1$

1. **Group the x's:** $7x - 3x = 4x$
2. **Group the constants:** $5 + 1 = 6$
3. **Reduced Form:** $4x + 6$